

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Styled Webpage with HTML5 Canvas Graphics and Flexbox Cards

2. AIM

To construct a modern styled webpage using CSS3 styling, linear gradient background, HTML5 Canvas 2D graphics drawing, and a responsive Flexbox card layout with hover animations.

3. OBJECTIVE

The objectives of this experiment are:

- Apply CSS3 styling rules including background gradients, box model borders, and rounded corners.
- Utilize HTML5 Canvas 2D context methods to draw geometric shapes and text.
- Build a flexbox card grid with elevation shadows and smooth hover transitions.

4. THEORY

CSS3 introduces advanced visual effects such as linear gradients (linear-gradient), box shadows (box-shadow), border radii (border-radius), and flexbox layouts.

HTML5 Canvas provides an API for drawing 2D graphics dynamically via JavaScript (fillRect, arc, fill, lineTo). Flexbox allows flexible element alignment across containers.

Application Flow:

Browser -> HTML5 Canvas -> 2D Context Drawing -> Flexbox Grid -> Rendered Webpage

5. ALGORITHM / PROCEDURE

1. Define CSS reset rules and linear-gradient background for body.
2. Style header and navigation bar with custom typography and hover state styling.
3. Add element with specified width and height.
4. Write JavaScript code to get 2D canvas context and render a blue rectangle, red circle, green line, and author name text.
5. Create container with flex display containing styled cards.
6. Apply CSS hover rules for interactive background and shadow effects.

6. PROGRAM

task2.html

```
<!DOCTYPE html>
<html>
<head>
```

```
<title>My Styled Webpage</title>

<style>
  * {
    margin: 0;
    padding: 0;
  }

  body {
    font-family: Arial, Helvetica, sans-serif;
    background: linear-gradient(white, #e8e8e8);
    padding: 15px;
  }

  header {
    text-align: center;
    padding: 10px;
    font-family: Georgia, serif;
  }

  header h1 {
    font-size: 32px;
  }

  header p {
    font-size: 16px;
    font-family: Arial, sans-serif;
  }

  nav {
    text-align: center;
    padding: 10px;
  }

  nav a {
    color: black;
    text-decoration: none;
    padding: 6px 16px;
    margin: 0 5px;
    border: 1px solid #ccc;
    border-radius: 5px;
  }

  nav a:hover {
    background: #ddd;
    color: #333;
    border-color: #999;
  }

  canvas {
    border: 2px solid #333;
    border-radius: 8px;
    display: block;
    margin: 10px auto;
  }

  h2 {
    text-align: center;
```

```
    font-family: Georgia, serif;
}

.container {
    display: flex;
    padding: 10px 0;
}

.card {
    border: 2px solid #666;
    border-radius: 10px;
    padding: 15px;
    width: 180px;
    margin: 5px;
    box-shadow: 3px 3px 8px #aaa;
}

.card h3 {
    font-size: 20px;
    font-family: Georgia, serif;
}

.card p {
    font-size: 14px;
    font-family: Arial, sans-serif;
}

.card:hover {
    background: #f5f5f5;
    box-shadow: 5px 5px 12px #888;
}

.card button {
    background: #444;
    color: white;
    border: 1px solid #666;
    border-radius: 5px;
    padding: 6px 22px;
    margin-top: 8px;
    box-shadow: 2px 2px 5px #aaa;
}

.card button:hover {
    background: #666;
    color: white;
    box-shadow: 3px 3px 8px #888;
}
</style>
</head>

<body>

    <header>
        <h1>My Styled Webpage</h1>
        <p>CSS3 Styling with HTML5 Canvas</p>
    </header>

    <nav>
```

```
<a href="#">Home</a>
<a href="#">About</a>
<a href="#">Services</a>
<a href="#">Contact</a>
</nav>

<hr>

<h2>Canvas Graphics</h2>

<canvas id="myCanvas" width="450" height="200"></canvas>

<script>
  var canvas = document.getElementById("myCanvas");
  var ctx = canvas.getContext("2d");

  ctx.fillStyle = "blue";
  ctx.fillRect(30, 30, 100, 60);
  ctx.fillStyle = "black";
  ctx.font = "12px Arial";
  ctx.fillText("Rectangle", 45, 110);

  ctx.beginPath();
  ctx.fillStyle = "red";
  ctx.arc(210, 60, 40, 0, 2 * Math.PI);
  ctx.fill();
  ctx.fillStyle = "black";
  ctx.fillText("Circle", 195, 120);

  ctx.beginPath();
  ctx.strokeStyle = "green";
  ctx.lineWidth = 4;
  ctx.moveTo(320, 30);
  ctx.lineTo(400, 90);
  ctx.stroke();
  ctx.fillStyle = "black";
  ctx.fillText("Line", 345, 120);

  ctx.fillStyle = "black";
  ctx.font = "bold 22px Arial";
  ctx.fillText("Harsha Vardhan", 150, 170);
</script>

<hr>

<div class="container">
  <div class="card">
    <h3>Web Dev</h3>
    <p>HTML, CSS, JavaScript</p>
    <button>Learn</button>
  </div>

  <div class="card">
    <h3>Design</h3>
    <p>Graphics & UI/UX</p>
    <button>Learn</button>
  </div>
</div>
```

```
<div class="card">
  <h3>Marketing</h3>
  <p>SEO & Social Media</p>
  <button>Learn</button>
</div>
</div>

</body>
</html>
```

7. CODE EXPLANATION

Code / Syntax	Explanation
background: linear-gradient(...);	Applies smooth multi-color linear gradient background to webpage.
display: flex;	Creates flexible flexbox layout container for card alignment.
.card:hover	Applies dynamic box-shadow and background color shift on cursor hover.
canvas.getContext('2d')	Obtains 2D rendering context for Canvas graphics operations.
ctx.fillRect(30, 30, 100, 60)	Draws filled blue rectangle on canvas.
ctx.arc(210, 60, 40, 0, 2*Math.PI)	Draws filled red circle shape.
ctx.fillText('Harsha Vardhan', ...)	Renders styled text directly onto the canvas.

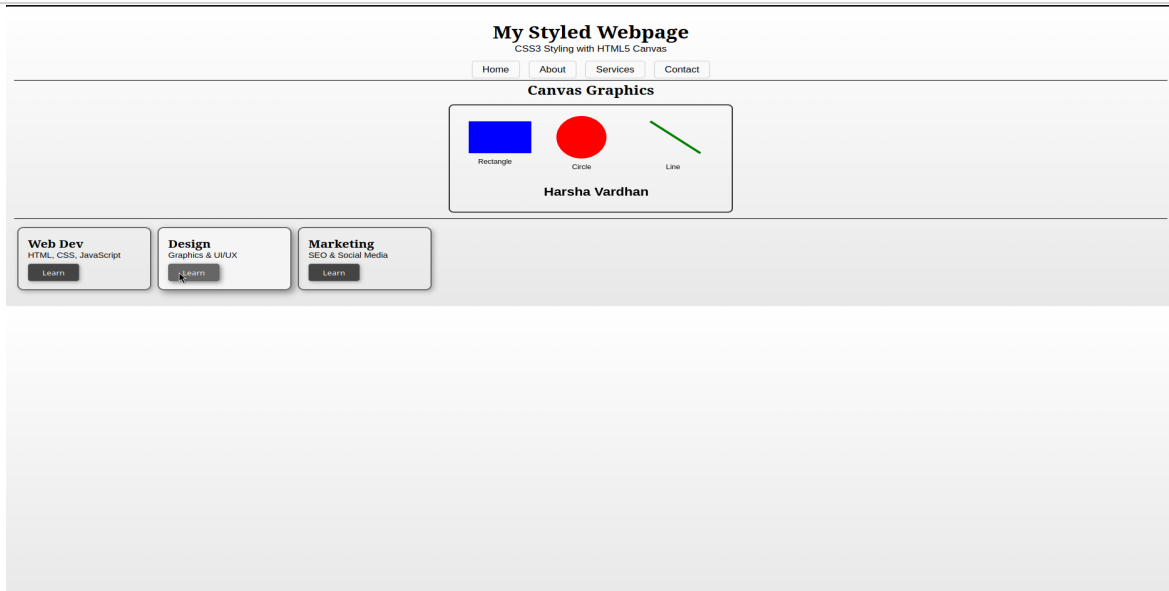
8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Page Gradient Background	Background gradient renders smoothly	Background gradient renders smoothly	Pass
TC-02	Navigation Bar Hover Effect	Nav buttons highlight on hover	Nav buttons highlight on hover	Pass
TC-03	Canvas Shapes Rendering	Canvas renders rectangle, circle, line, and text	Canvas renders rectangle, circle, line, and text	Pass
TC-04	Flexbox Card Grid Layout	Cards align horizontally with spacing	Cards align horizontally with spacing	Pass
TC-05	Card Hover Elevation	Card shadow expands on mouse hover	Card shadow expands on mouse hover	Pass

9. OUTPUT

Output 1: Canvas Graphics & Flexbox Card Grid Output



10. RESULT

Thus, the Styled Webpage incorporating HTML5 Canvas 2D graphics and CSS3 Flexbox card components was successfully developed and verified across all test cases.